C#_Week4_Day1_Hands_On -- Kaveri Harish Babu

C#

Level-1 Problem 1: Student Grade Evaluator

Scenario

You are developing a console-based application in .NET 8 for a school. The application should evaluate a student's marks and assign a grade based on predefined rules.

Requirements

- Accept student name and marks (0-100).
- Use if-else statements to determine grade.
- Display grade as A, B, C, D or Fail.
- Handle invalid input using conditional checks.

Technical Constraints

- Use C# (.NET 8 Console Application).
- Use appropriate data types (string, int).
- Use if-else control flow.
- Do not use advanced concepts like classes or LINQ.

Sample Input

Enter Name: Rahul
Enter Marks: 78

Sample Output

Student: Rahul
Grade: B

Expectations

Program should correctly evaluate grade and handle edge cases like marks below 0 or above 100.

Learning Outcome

Understand variables, data types, input/output handling and if-else control statements in C#.

Source Code:

File Name: Student_Grade_Evaluator.cs

```
using System;
using System.Collections.Generic;
using System.Text;

namespace ConsoleApp1
{
    0 references
    internal class Student_Grade_Evaluator
    {
        0 references
        static void Main()
        {
            Console.Write("Enter your Name: ");
            string name = Console.ReadLine();
            if (string.IsNullOrWhiteSpace(name))
            {
                Console.WriteLine("Please Enter a Valid Student Name.");
            }
            else
            {

                Console.Write("Enter your marks: ");
                int marks = Convert.ToInt32(Console.ReadLine());

                if (marks < 0 || marks > 100)
                {
                    Console.WriteLine("Invalid Marks. Please Enter the Marks (1-100)");
                }
                else if (marks >= 90)
                {
                    Console.WriteLine("Student: " + name);
                    Console.WriteLine("Grade A");
                }
                else if (marks >= 70)
                {
                    Console.WriteLine("Student: " + name);
```

```
                if (marks < 0 || marks > 100)
                {
                    Console.WriteLine("Invalid Marks. Please Enter the Marks (1-100)");
                }
                else if (marks >= 90)
                {
                    Console.WriteLine("Student: " + name);
                    Console.WriteLine("Grade A");
                }
                else if (marks >= 70)
                {
                    Console.WriteLine("Student: " + name);
                    Console.WriteLine("Grade B");
                }
                else if (marks >= 50)
                {
                    Console.WriteLine("Student: " + name);
                    Console.WriteLine("Grade C");
                }
                else if (marks >= 35)
                {
                    Console.WriteLine("Student: " + name);
                    Console.WriteLine("Grade D");
                }
                else
                {
                    Console.WriteLine("Student: " + name);
                    Console.WriteLine("Fail");
                }
            }
        }
    }
}
```

Output:

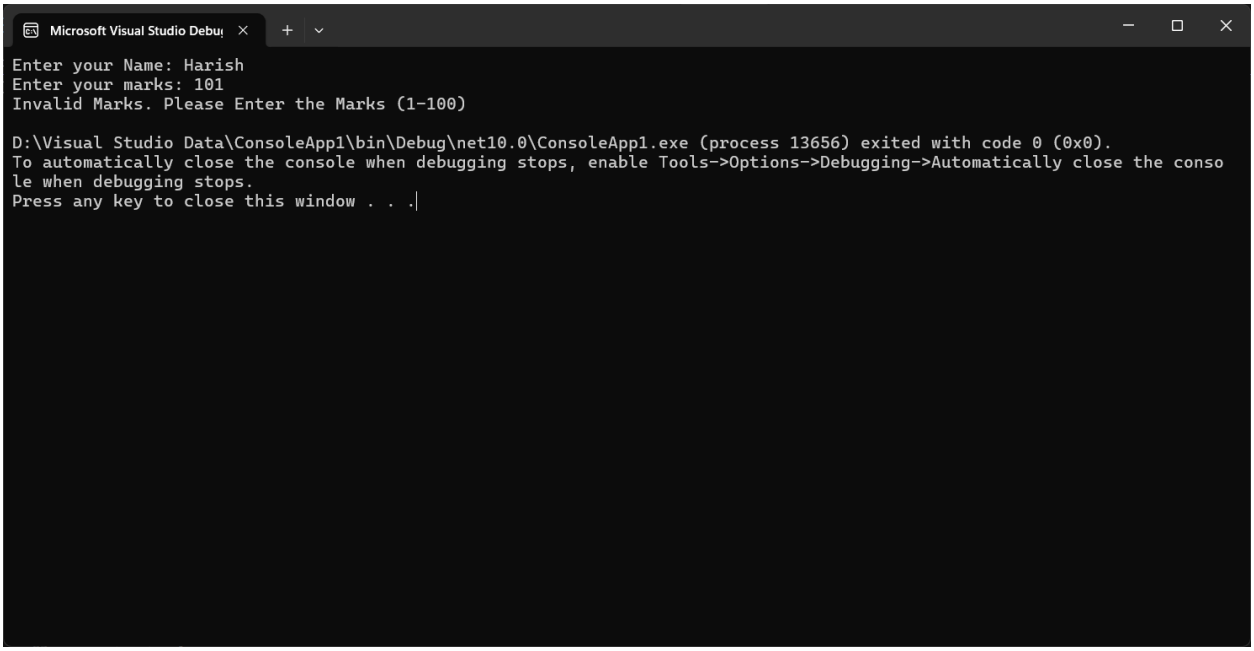
Output File: Student_Grade_Evaluator.cs

```
Microsoft Visual Studio Debu
Enter your Name: Harish
Enter your marks: 78
Student: Harish
Grade B

D:\Visual Studio Data\ConsoleApp1\bin\Debug\net10.0\ConsoleApp1.exe (process 23504) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .
```

```
Microsoft Visual Studio Debu
Enter your Name:
Please Enter a Valid Student Name.

D:\Visual Studio Data\ConsoleApp1\bin\Debug\net10.0\ConsoleApp1.exe (process 23764) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .
```



Explanation:

This C# program asks the user to enter a **student name and marks**, then checks if the input is valid. Based on the marks, it uses **if-else conditions** to assign a grade like **A, B, C, D, or Fail**. The output shows the **student name and grade**, and it also displays error messages if the name is empty or the marks are outside **0–100**.

Level-1 Problem 2: Simple Calculator Using Switch

Scenario

Create a simple calculator application that performs basic arithmetic operations.

Requirements

- Accept two numbers from user.
- Accept operator (+, -, *, /).
- Use switch statement to perform operation.
- Display result.

Technical Constraints

- Use int or double data types.
- Use switch-case statement.
- Handle division by zero.

Sample Input

Enter First Number: 10
Enter Second Number: 5
Enter Operator: *

Sample Output

Result: 50

Expectations

Correct operator selection and proper validation of inputs.

Learning Outcome

Understand switch statements, arithmetic operators and control flow in C#.

Source Code:

File Name: Simple_Calculator_Using_Switch.cs

```
using System;
using System.Collections.Generic;
using System.Text;

namespace ConsoleApp1
{

    internal class Simple_Calculator_Using_Switch
    {
        static void Main()
        {
            Console.Write("Enter First Number: ");
            double num1 = Convert.ToDouble(Console.ReadLine());

            Console.Write("Enter Second Number: ");
            double num2 = Convert.ToDouble(Console.ReadLine());

            Console.Write("Enter Operator (+, -, *, /): ");
            char operators = Convert.ToChar(Console.ReadLine());

            double result = 0;

            switch (operators)
            {
                case '+':
                    result = num1 + num2;
                    Console.WriteLine("Result: " + result);
                    break;

                case '-':
                    result = num1 - num2;
                    Console.WriteLine("Result: " + result);
                    break;
```

```
            {
                case '+':
                    result = num1 + num2;
                    Console.WriteLine("Result: " + result);
                    break;

                case '-':
                    result = num1 - num2;
                    Console.WriteLine("Result: " + result);
                    break;

                case '*':
                    result = num1 * num2;
                    Console.WriteLine("Result: " + result);
                    break;

                case '/':
                    if (num2 == 0)
                    {
                        Console.WriteLine("Error: Division by zero is not allowed.");
                    }
                    else
                    {
                        result = num1 / num2;
                        Console.WriteLine("Result: " + result);
                    }
                    break;

                default:
                    Console.WriteLine("Invalid Operator!");
                    break;
            }
        }
    }
}
```

Output:

Output File: Simple_Calculator_Using_Switch.cs

```
Microsoft Visual Studio Debu

Enter First Number: 73
Enter Second Number: 50
Enter Operator (+, -, *, /): +
Result: 123

D:\Visual Studio Data\ConsoleApp1\bin\Debug\net10.0\ConsoleApp1.exe (process 1180) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .
```

```
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Enter First Number: 50
Enter Second Number: 70
Enter Operator (+, -, *, /): -
Result: -20

D:\Visual Studio Data\ConsoleApp1\bin\Debug\net10.0\ConsoleApp1.exe (process 27000) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .
```

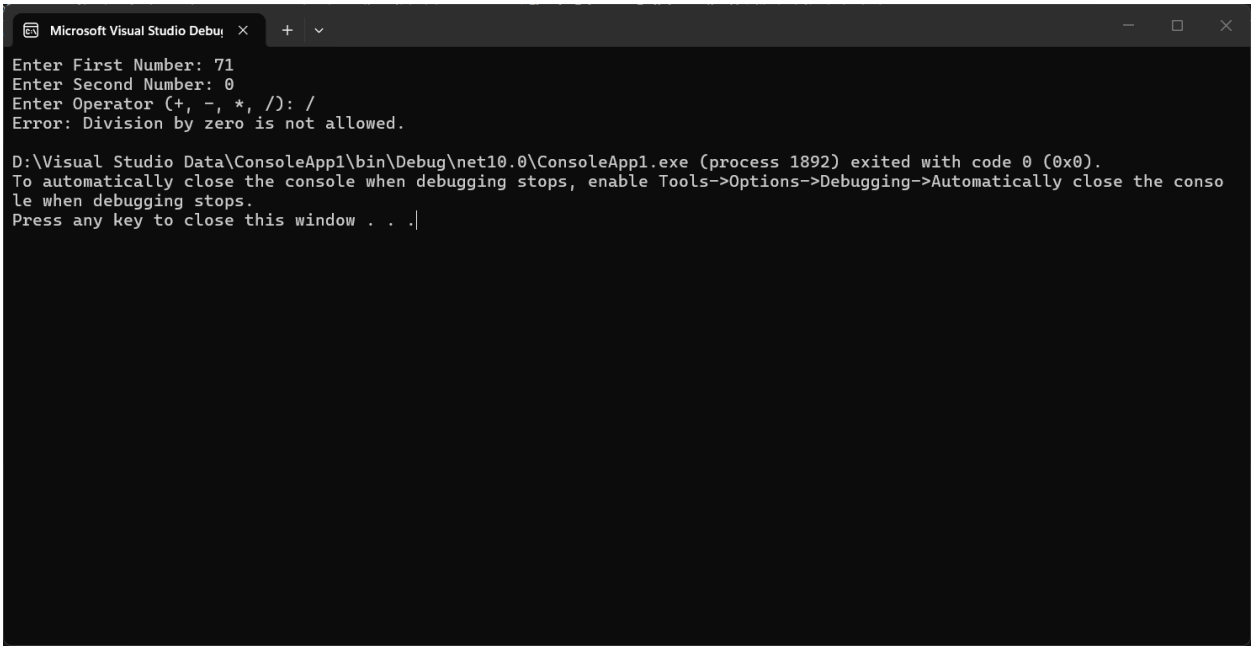

Explanation:

This C# program is a **simple calculator** that asks the user to enter two numbers and an operator (+, −, *, /). It uses a **switch statement** to perform the selected arithmetic operation. The output shows the **calculated result**, and if the user tries to divide by zero, it displays an **error message saying division by zero is not allowed**.

Level-2 Problem 1: Employee Bonus Calculator

Scenario

Develop a console application that calculates employee bonus based on salary and years of experience.

Requirements

- Accept employee name, salary and years of experience.
- Use if-else and conditional operator.
- Bonus rules:
  - Experience < 2 years: 5% bonus
  - 2-5 years: 10% bonus
  - >5 years: 15% bonus
- Display final salary after bonus.

Technical Constraints

- Use double for salary.
- Use if-else and ternary operator.
- Use proper formatting for currency output.

Sample Input

Enter Name: Aisha
Enter Salary: 50000
Enter Experience: 4

Sample Output

Employee: Aisha
Bonus: 5000
Final Salary: 55000

Expectations

Accurate bonus calculation and correct usage of control statements.

Learning Outcome

Apply conditional logic and arithmetic operations in real-world scenarios.

Source Code:

File Name: Employee_Bonus_Calculator.cs

```
using System;
using System.Collections.Generic;
using System.Text;

namespace ConsoleApp1
{
    0 references
    internal class Employee_Bonus_Calculator
    {
        0 references
        static void Main()
        {
            Console.Write("Enter Name: ");
            string name = Console.ReadLine();

            Console.Write("Enter Salary: ");
            int salary = Convert.ToInt32(Console.ReadLine());

            Console.Write("Enter Experience: ");
            int experience = Convert.ToInt32(Console.ReadLine());

            double bonusPercent;

            if (experience < 2)
            {
                bonusPercent = 0.05;
            }
            else if (experience <= 5)
            {
                bonusPercent = 0.1;
            }
            else
            {
                bonusPercent = 0.15;
            }
```

```
            Console.Write("Enter Experience: ");
            int experience = Convert.ToInt32(Console.ReadLine());

            double bonusPercent;

            if (experience < 2)
            {
                bonusPercent = 0.05;
            }
            else if (experience <= 5)
            {
                bonusPercent = 0.1;
            }
            else
            {
                bonusPercent = 0.15;
            }

            double bonus = salary * bonusPercent;

            double FinalSalary = salary + bonus;

            Console.WriteLine("Employee: " + name);
            Console.WriteLine("Bonus: " + bonus);
            Console.WriteLine("Final Salary: " + FinalSalary);
        }
    }
}
```

Output:

Output File: Employee_Bonus_Calculator.cs

```
Enter Name: Aisha
Enter Salary: 50000
Enter Experience: 4
Employee: Aisha
Bonus: 5000
Final Salary: 55000

D:\Visual Studio Data\ConsoleApp1\bin\Debug\net10.0\ConsoleApp1.exe (process 27284) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .
```

```
Enter Name: Harish
Enter Salary: 100000
Enter Experience: 6
Employee: Harish
Bonus: 15000
Final Salary: 115000

D:\Visual Studio Data\ConsoleApp1\bin\Debug\net10.0\ConsoleApp1.exe (process 14024) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .
```

Explanation:

This C# program asks the user to enter the **employee name, salary, and years of experience**. Based on the experience, it calculates the **bonus percentage (5%, 10%, or 15%)** using if-else conditions. The output shows the **employee name, calculated bonus amount, and the final salary after adding the bonus**.

Level-2 Problem 2: Number Analysis Using Loops

Scenario

Create a .NET 8 console application that analyzes numbers between 1 and N.

Requirements

- Accept a number N from user.
- Use loops to:
  - Count even numbers
  - Count odd numbers
  - Calculate sum of all numbers
- Display results.

Technical Constraints

- Use for or while loop.
- Use int data type.
- Avoid using arrays or collections.

Sample Input

Enter Number: 10

Sample Output

Even Count: 5
Odd Count: 5
Sum: 55

Expectations

Proper loop usage and correct counting logic.

Learning Outcome

Strengthen understanding of loops, counters and accumulators in C#.

Source Code:

File Name: Number_Analysis_Using_Loops.cs

```
using System;
using System.Collections.Generic;
using System.Text;

namespace ConsoleApp1
{
    0 references
    internal class Number_Analysis_Using_Loops
    {
        0 references
        static void Main()
        {
            Console.Write("Enter a Number: ");
            int n = Convert.ToInt32(Console.ReadLine());

            int Even_Count = 0;
            int Odd_Count = 0;
            int sum = 0;

            for (int i = 1; i <= n; i++)
            {
                sum +=  i;

                if(i%2==0)
                {
                    Even_Count++;
                }
                else
                {
                    Odd_Count++;
                }
            }
            Console.WriteLine("Even Count: " + Even_Count);
            Console.WriteLine("Odd Count: " + Odd_Count);
            Console.WriteLine("Sum: " + sum);
        }
    }
}
```

Output:

Output File: Number_Analysis_Using_Loops.cs

```
Enter a Number: 10
Even Count: 5
Odd Count: 5
Sum: 55

D:\Visual Studio Data\ConsoleApp1\bin\Debug\net10.0\ConsoleApp1.exe (process 10756) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .
```

```
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Enter a Number: 111
Even Count: 55
Odd Count: 56
Sum: 6216

D:\Visual Studio Data\ConsoleApp1\bin\Debug\net10.0\ConsoleApp1.exe (process 17244) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .
```

Explanation:

This C# program asks the user to enter a number **n** and then uses a **for loop** to check numbers from **1 to n**. It counts how many numbers are **even and odd** and also calculates the **sum of all numbers** in that range. The output shows the **total even count, odd count, and the sum of numbers from 1 to n**.